Be warned, though, that in figure 17.1 we are not showing all the failed attempts where local random optimization tried the wrong direction: overall, getting to the same point will need more steps than gradient descent, because instead of going in the direction of maximum change, we are wandering around randomly: this is more apparent looking at a 2D domain/ 3Dsurface, like the one shown in figure 17.2.

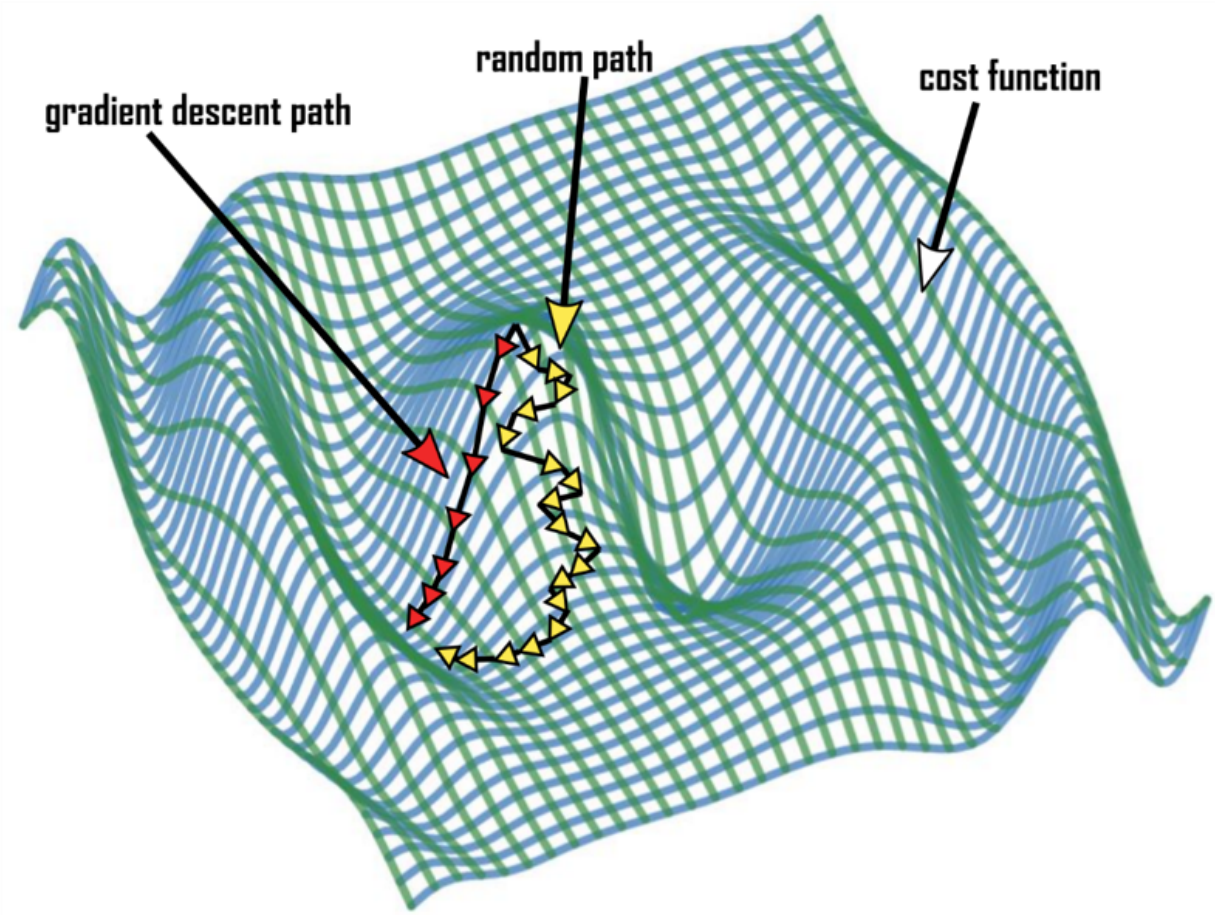


Figure 17.2 Gradient descent versus local random optimization on a cost function of two parameters. Can you guess which one is going to take more steps?

As it is, this approach shares some of the same issues we discussed in chapter 16 about gradient descent: it is likely to